JIALIN LI

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EDUCATION & ACADEMIC EMPLOYMENT

Department of Math & Statistics, University of Massachusetts, Amherst 2024–Present
Visiting Assistant Professor

Institute for Mathematical and Statistical Innovation, University of Chicago Spring 2025
Visiting Research Associate

Rotman School of Management, University of Toronto 2021–2024
Postdoctoral Research Fellow

University of Maryland, College Park 2016–2021
Ph.D., Applied Mathematics & Statistics, and Scientific Computation

Nankai University 2012–2016
B.S., Statistics

RESEARCH INTERESTS

- Theory: uncertainty quantification including Gaussian processes and nonparametric methods.
- Methodology: fundamental methods for system intelligence and industrial computing, where systems are expensive to observe and only partially understood
- Applications with a social-good theme: privacy, service quality, environment, resilient agriculture

WORKING PAPERS

- Jialin Li, Nonnegativity shapes predictive uncertainty: Latent process factors and quantile construction. To be submitted soon. Link
  - To be presented at INFORMS Annual Meeting 2026: Session TA28; Time: Tuesday, November 3, 8:50 AM–9:15 AM; Location: Moscone South-206 (Level 2).
- Jialin Li, Zihao Qu, and Mengfan Xu, Dynamic balancing and matchmaking for engagement management. Major revision at *Management Science*. Link
- Ningyuan Chen, Ming Hu, Jialin Li, and Sheng Liu, Data privacy in pricing: Estimation bias and implications. Resubmitted after minor revision at *Manufacturing & Service Operations Management*. Link
  - MSOM Conference 2026 the Interface of Finance, Operations and Risk Management (iFORM) Special Interest Group (SIG) Day Presentation.
- Furong Huang, Jialin Li, and Xuchen You, Guaranteed simultaneous asymmetric tensor decomposition via alternating subspace iteration. Available on arXiv. Link

PUBLICATIONS

- Jialin Li and Ilya O. Ryzhov (2024), Moderate deviations inequalities for Gaussian process regression. *Journal of Applied Probability* 61(1): 172–197. Link
  - Selected for inclusion in the February 2025 Applied Probability Collection on the topic of Gaussian processes. This collection features ten notable articles published by the Applied Probability Trust. Link
- Jialin Li and Ilya O. Ryzhov (2022), Convergence rates of epsilon-greedy global optimization under radial basis function interpolation. *Stochastic Systems* 13(1): 59–92. Link

INDUSTRY EXPERIENCE

Data Scientist Intern, Google Summer 2021

- Improved the performance of the A/B test model for analyzing the heterogeneous impact of launched features on the advertisement revenue or impressions of YouTube creators' channels
- Developed score-based methods for model selection based on real experiment data

Quantitative Intern, Wells Fargo Summer 2020

- Audited several asset-pricing models and reviewed technical reports on model development, model parameter calibration, model validation, and performance monitoring plans
- Assisted model development and model validation teams and completed a report to provide technical opinions

TEACHING EXPERIENCE

Course Instructor, University of Massachusetts, Amherst

- Linear Regression and Analysis of Variance (Fall 2025, Spring 2026, Fall 2026)
- Statistics I (Fall 2024, Fall 2025, Spring 2026)

Course Instructor, University of Maryland, College Park

- Introduction to Math Modeling and Probability (Spring 2017)

Teaching Assistant, University of Maryland, College Park

- Undergraduate Course Discussion Session Instructor (to give lectures and lead Q&A discussions in the complementary sessions for regular course sessions): Linear Algebra (Spring 2021), Applied Probability and Statistics (Spring & Fall 2020), Calculus III (Fall 2018), Linear Algebra for Scientists and Engineers (Spring 2018, Fall 2019), Calculus II (Fall 2017)
- Ph.D. Course Grader: Stochastic Optimization (Spring 2020), Stochastic Process (Spring 2019), Linear Model (Spring 2019)

PRESENTATIONS

- "Non-Negativity Shapes Predictive Uncertainty: Latent Gaussian Process Factors and Quantile Construction," INFORMS Annual Meeting 2026 (scheduled)
- "Data Privacy in Pricing: Estimation Bias and Implications," MSOM Conference 2026 SIG Presentation
- "Probabilistic Non-Negative Factorization with Latent Gaussian Processes," INFORMS Annual Meeting 2025
- "Randomized Experimental Design and New Convergence Rate Results in Black-Box Optimization with RBF Surrogates," IMSI Long Program: Uncertainty Quantification and AI for Complex Systems, 2025
- "Convergence Rate Results in Black-Box Optimization with Surrogate Models," Statistics and Data Science Seminar, UMass Amherst, 2024
- "Data Privacy in Pricing: Estimation Bias and Implications," INFORMS Annual Meeting 2024
- "Epsilon-Greedy Global Optimization Under Radial Basis Function Interpolation," Learning Learning Seminar, UMass Amherst, 2024
- "Data Privacy in Pricing: Estimation Bias and Implications," Rotman TD Management Data and Analytics Lab Grant Research Roundtable, 2024
- "Data Privacy in Pricing: Estimation Bias and Implications," INFORMS Annual Meeting 2023
- "Data Privacy in Pricing: Estimation Bias and Implications," MSOM Conference 2023
- "Pricing Under Privacy Protection," INFORMS Annual Meeting 2022
- "Incentivizing Myopic Customers to Explore," CORS/INFORMS International Conference 2022
- "Moderate Deviations Inequalities for Gaussian Process Regression," INFORMS Annual Meeting 2021
- "Epsilon-greedy Global Optimization Under Radial Basis Function Interpolation," Rotman Young Scholar Seminar, 2021
- "Convergence Rates of Global Optimization Under Randomized Sampling," INFORMS Annual Meeting 2019

FUNDING & AWARDS

- MSP Research Support Fund, UMass Amherst, 2025 & 2026
- TD Management Data and Analytics Lab Research Grant, Rotman School of Management, 2023
- New Pilot Postdoc Funding, Rotman School of Management, 2022
- Graduate Student Summer Research Fellowship, University of Maryland, 2019

ACADEMIC SERVICE

Ad hoc Reviewer

- Operations Research, Statistica Sinica, IISE Transactions, SIAM/ASA Journal on Uncertainty Quantification (JUQ), Operations Research Letters (ORL), Winter Simulation Conference (WSC), IISE Annual Conference, International Conference on Machine Learning (ICML), International Joint Conference on Artificial Intelligence (IJCAI), European Conference on Artificial Intelligence (ECAI)